

RESEARCH ARTICLE

Fourteen-Year Fluvial Sediment Record Shows Non-Conservativeness of Organic Tracers: Recommendations for Sediment Fingerprinting

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Received: 29 October 2024 | **Revised:** 19 December 2024 | **Accepted:** 23 December 2024

Funding: Kentucky Senate Bill 271B Water Quality program and National Science Foundation Awards #1632888, #2119853 and #1933779.

Keywords: carbon | erosion | isotopic ratio | nitrogen | sediment sources | sediment tracing

ABSTRACT

While tracing the sources of fluvial sediment using carbon and nitrogen stable isotopic ratios ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) has progressed significantly over the last two decades, the conservativeness of these tracers remains questionable. Recent work indicates that $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ alterations in streambed deposition zones likely represent the largest source of uncertainty impacting usefulness of the isotopic ratios as tracers. Here we report a 14-year dataset of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of fluvial sediment from a streambed-dominated basin in Kentucky, USA, and employ empirical model decomposition (EMD) to identify dominant temporal trends that may impact conservativeness. Results from EMD show significant seasonality of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ for sediment as well as underlying multi-year variation. The seasonal and multi-year variance account for 72% and 50% of the total data variation for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively. The prominent seasonality for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ show a mean intra-annual change of 0.6‰ and 1.1‰, respectively, and the seasonal change is attributed to algal accrual and organic matter turnover in the streambed sediment deposits. Mixing model simulations show that the mean streambed isotopic ratios should be separated from other sediment sources by 3.0‰ and 3.6‰ for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively, to achieve 90% accuracy in source apportionment when the isotopic ratios are used independently; and the mean streambed value of both isotopic ratios should be separated from other sediment sources by 3.0‰ when $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ are used in combination. Our results lead to the recommendation that isotope ratios of sources be separated by at least 3‰ when the streambed is expected to be a prominent sediment source, which far exceeds the prior recommendation of 1‰ mean separation of sources.

1 | Introduction

Over the last two decades sediment fingerprinting using carbon and nitrogen stable isotopic ratios ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) as tracers has been increasingly used in watershed sedimentation studies. These studies show the progression of the method, from a proof of concept to discriminate sediment sources (Papanicolaou, Fox, and Marshall 2003) to a variety of research applications, such as

assessing soil conservation measures (Zhang et al. 2017; Collins et al. 2019); assessing sediment carbon release dynamics in soils and export in watersheds (Bellanger et al. 2004; Dalzell, Filley, and Harbor 2007); studying the impact of land use changes, such as reforestation, on soil erosion (Huang et al. 2017); and apportioning organic matter from specific vegetation cover with compound-specific isotope analyses (Gibbs 2008; Blake et al. 2012; Cooper et al. 2015).

The applicability of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ to discriminate land use, streambank and streambed sources in general reflects the plant growth at the source, organic matter degradation and nitrogen cycling. $\delta^{13}\text{C}$ may discriminate soils from different land uses because the sediment tracer reflects the distribution of C3 and C4 plants growing on the landscape with some isotopic change during degradation (Smith and Epstein 1971; Farquhar, Ehleringer, and Hubick 1989; Palmer et al. 2001; Jacinthe, Lal, and Owens 2009). $\delta^{15}\text{N}$ discriminates soils from different land uses as function of management methods (e.g., tillage, fertilisation) that impact nitrogen sources and transfers with isotopic changes in often predictable and recognisable directions (Kendall, Elliott, and Wankel 2007; Sharp 2017). Both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ generally increase with depth in soils as a result of isotope fractionation during organic matter turnover (e.g., Natelhoffer and Fry 1988; Campbell et al. 2009) and therefore the isotopic ratios can be useful for discriminating exposed streambank sources (Fox 2005; Mukundan et al. 2010). $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ can potentially discriminate deposited streambed sources from other sources because the sediment accrues the autochthonous growth of biomass, which can provide unique isotopic ratios in some systems (Ford et al. 2015; Ford and Fox 2017). Given the dependence of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ on organic matter in sediment, application of stable isotope composition amongst individual organic compounds is increasingly applied.

Based on review of the published studies, we found that a literature review of applying $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ tracers of sediment has not been recently published. Therefore, a first focus of this contribution reviews studies that use $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ as tracers in sediment fingerprinting studies. The review will include study type, spatial and temporal scale of the applications, and sediment source types and considerations. Thereafter, emphasis of the review is placed on treatment of tracer conservativeness for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, including methods applied to account for conservativeness, studies explicitly investigating conservativeness and recommendations to account for conservativeness.

Previously reported results suggest that the least resolved process impacting tracer conservativeness is uncertainty associated with changes to $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of the streambed sediment source (Riddle et al. 2022). Given this open line of investigation,

the second objective of the present contribution focuses on estimating changes to $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ tracers in streambed deposition zones in rivers and accounting for the potential non-conservativeness of tracers in source apportionment.

The existing theory of sediment dynamics and biogeochemistry suggests mechanisms that potentially impact conservativeness of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ tracers (Figure 1). Transported sediment fall out of suspension and deposit to the bed of the stream when the fluid's transport capacity to carry sediment eroded from sources, such as of the land surface via rill, interrill or gully erosion or from the streambanks, is insufficient relative to the sediment load (Yalin 1972; Graf 1984; Chien and Wan 1999). The fluvial sediment can accrue in streambed deposits (Figure 1) and thereafter assimilate as media for the stream's benthic ecosystem (Teal, Parker, and Solan 2010; Hope, Paterson, and Thrush 2020). Autochthonous growth of benthic algae and aquatic macrophytes and subsequent microbially-mediated degradation can lead to accrual of aquatic organic matter remnants in streambed sediment (Lara and Thomas 1995; Leloup et al. 2013; Hotchkiss and Hall Jr 2015). These processes can change the sediment organic matter makeup and $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of the streambed deposits—algae for example often show $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ that differs from upland and streambank sources (Ford, Fox, and Rowe 2015; Ford and Fox 2017).

Since the benthic ecosystem is continuously cycling between autotrophic and heterotrophic dominance, the potential exists for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of streambed sediment to be under continuous change. In this manner, upland and bank sediment sources that temporarily reside in the streambed can accrue algae, and the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ tracers may behave non-conservatively (Fox, Davis, and Martin 2010; Riddle et al. 2022). Likewise, treating streambed sediment as a unique source in sediment fingerprinting poses the potential for non-stationary tracers in which $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ are continually changing (Mahoney et al. 2019; Ford et al. 2020). These definitions of non-conservative isotopic tracers for temporary deposits and the existence of non-stationary tracers for the streambed source reflect the same process in the river, with their ambiguity and operational definition ultimately the decision of the scientist leading the fingerprinting study.

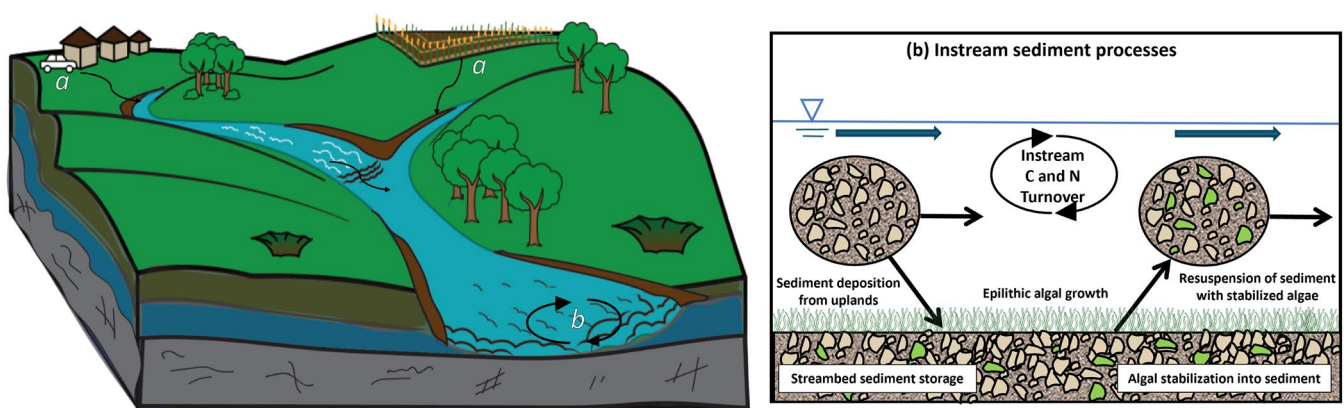


FIGURE 1 | Concept of low-gradient stream system in mixed urban and agricultural land use watershed. (a) Upland soil is eroded and connected to stream corridor during storm events. (b) Soil and particulate organic matter transported and deposited to streambed where intermittent organic matter turnover and algal growth stabilise to instream deposits.

Herein, we refer to the process as conservativeness of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ for streambed sediment.

Empirical mode decomposition provides a data-driven approach to assess significance of oscillations at varying temporal-scales (Huang et al. 1998), enabling investigation of the conservativeness of isotopic $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ and elemental tracers that may be continually changing in streambed sediment. A key advantage of EMD over other time-series analysis is that it does not assume parametric, linear or stationary characteristics of datasets (Huang et al. 1998; Wu et al. 2007). This is important given findings from previous physically based modelling efforts have demonstrated that both elemental and isotopic composition of streambed sediments may be non-stationary through time due to autochthonous accrual (Jensen et al. 2018), as well as abrupt shifts in composition due to extreme hydrologic events (Ford and Fox 2014a). Likewise, sediment elemental compositions are recognised to be non-normally distributed due to this non-conservative behaviour (Ford and Fox 2014b).

Our objectives were (i) to review the sediment tracers using $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ to date, and (ii) to estimate the time-varying signature of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of streambed sediment using empirical model decomposition and consider the tracer's potential non-conservativeness for the application of sediment fingerprinting. The first part is accomplished in the next section, and focuses on reviewing sediment sources as well as conservativeness of the tracers. To achieve the second objective, we collected and analysed a dataset from a watershed dominated by the streambed sediment source located in central Kentucky, United States. The data included 14 years of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ measurements from two sediment traps collected weekly to bi-weekly, which is one contribution of this paper considering these river datasets of isotopic ratios are typically much shorter in duration. The transported sediment collected with the sediment trap samples were 85% originated from streambed sediment, which allowed investigation of the time varying signature of the streambed. Empirical mode decomposition was used to separate the time-series of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ to the statistically significant long term mean and seasonal intrinsic mode functions (IMFs) as well as statistical noise. We discuss the likely reasons for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ non-conservativeness of the streambed sediment, and statistically analyse distribution of data noise. We carry out mixing model simulations to assist with providing recommendations for sediment fingerprinting with $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, when the streambed sediment sources is expected to be a major source to the rivers' sediment load. Finally, we carry out a number of analyses to consider the robustness of our results when analysing the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ time series.

2 | Literature Review

We found 49 studies across five continents and two decades, that use $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ to apportion sediment sources (Table 1). These studies applied the isotopic ratios to discriminate land use sources of sediment (12 studies), land use as well as streambank sources (21 studies), and increasingly studies also discriminate streambed sources (16 studies). The studies are varied in scale with the watershed size varying from the less than one square

kilometre field scale (e.g., Collins et al. 2019) to nearly 2 million square kilometre basin scale (Sun et al. 2021). Field sampling of transported sediment varies widely as well from sampling a storm event on a single occasion (Upadhayay et al. 2018; Bahadori et al. 2019) to more than a decade of data collection (Tian et al. 2019). Sediment sources vary widely to include soil from different agricultural and forest practises, bank soils and subsoils, spoil from mining and construction activities, sediment from roads and artificial levees, and sediment from streambed deposits (Table 1).

Regarding conservativeness of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ as tracers in fingerprinting, many studies discuss and attempt to consider conservativeness as well as bring attention to the matter, albeit often highlighting tracer conservativeness as an underlying assumption requiring more investigation. 80% of reported studies (39 out of 49 studies, Table 1) carry out concentration dependent mixing when using $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$. Concentration dependent mixing is physically necessary via mass balance because $\delta^{13}\text{C}$ mixes carbon and $\delta^{15}\text{N}$ mixes nitrogen, not sediment (e.g., derivation detailed in Fox and Martin 2015). The discrepancies in the density (i.e., concentration) of carbon and nitrogen of sources and transported sediment in rivers occurs primarily due to loss of organic matter during disaggregation and transport of fluvial sediment (Riddle et al. 2022), although an increase in organic matter content is not impossible in the case of particle size sorting during transport. In this manner, concentration dependent mixing accounts for physical impacts on $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ conservativeness.

86% of reported studies (42 of 49 studies, Table 1) carry out a bracket test to consider conservativeness of the isotopic ratios, and the bracket test is also referred to as the range test or conservation test in some studies. The bracket test accepts the isotopic ratio tracers (or any tracers, for that matter) falling in the range comprised by the sediment source distribution for the tracers; and rejects tracers outside of the sediment source distribution. 82% of studies found $\delta^{13}\text{C}$ passed the bracket test with 32 studies finding that $\delta^{13}\text{C}$ passed while 7 studies found that $\delta^{13}\text{C}$ failed the bracket test. 78% of studies found $\delta^{15}\text{N}$ passed the bracket test with 25 studies finding that $\delta^{15}\text{N}$ passed while 7 studies found that $\delta^{15}\text{N}$ failed the bracket test. Not all studies measured or considered both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, and seven studies either did not consider the bracket test or did not mention carrying out the method. An approximate 80% success rate for the isotopic ratios offers some optimism regarding their conservativeness to some degree. However, we point out that $\delta^{13}\text{C}$ or $\delta^{15}\text{N}$ could change by one or more per mil and still be 'in-bounds', depending on the overall range of the sources, which in turn would change apportionment results.

Some studies explicitly address conservativeness considerations in their study. Fox, Davis, and Martin (2010) carried out biogeochemical modelling of temporary stored sediment and showed the potential for $\delta^{15}\text{N}$ changes due to nitrogen cycling. Garzon-Garcia et al. (2017) accounted for conservativeness during sediment transport by preparing source samples with methods to mimic mobilisation. Collins et al. (2019) measured $\delta^{13}\text{C}$ or $\delta^{15}\text{N}$ of soils from farmlands and showed measurable and significant shifts due to fertilisation. Mahoney et al. (2019) used time series decomposition methods to account for $\delta^{13}\text{C}$ or $\delta^{15}\text{N}$ changes during storage. Ford et al. (2020) parameterized

TABLE 1 | Literature review of sediment fingerprinting using C and N stable isotopic ratios.

Study	Location	Watershed (km ²)	Sampling duration	Sediment sources	Tracers	Conc. dependent mixing	Bracket/range test or other test (P =pass; F =fail test)
Studies considering uplands with different land uses and vegetative cover							
Papanicolaou, Fox, and Marshall 2003	Palouse River, Washington, USA	600	1 month	Winter wheat agriculture and conifer forest silt loam soil	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	NO	Performed: NO
Bellanger et al. 2004	Bocono, Venezuela	25**	5 months	Experimental coffee, maize, and bare plots	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: NO
Dalzell, Filley, and Harbor 2007	Big Pine Creek, Indiana, USA	850	1 year	Corn, soybean, tall fescue	$\delta^{13}\text{C}$	NO	Performed: NO
Gibbs 2008	Mahurangi Harbour, New Zealand	117	1 month	Pastures, forested areas	$\delta^{13}\text{C}$ CSIA	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: Not measured
Jacinthe, Lal, and Owens 2009	North Appalachian Experimental Watershed, Ohio, USA	Unspecified	2 month	Corn, soy bean, rye rotation, ploughed continuous corn, no-till	$\delta^{13}\text{C}$	YES	Performed: NO
Huang et al. 2017	Dianchi watershed. Southwest China	2800	1 month	Forests, farmland	$\delta^{13}\text{C}$	YES	Performed: YES $\delta^{13}\text{C}$: F $\delta^{15}\text{N}$: Not measured
Liu et al. 2018	Xijiazhai watershed, China	3.1	2 months	Forest, grassland, fallow, gully, cropland	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Upadhayay et al. 2018	Susindol, Nepal; Enziwigger, Central Switzerland	1, 31	1 occasion	Mixed forest, pine forest, and terraced agricultural upland	$\delta^{13}\text{C}$ CSIA	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: Not measured
Bahadori et al. 2019	Johnstone River catchment, Australia	2624	1 occasion	Grazing (pasture), sugar cane, forest, banana	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Collins et al. 2019	North Wyke Farm Platform, Okehampton, Devon, UK	0.018	2 months	Topsoil, subsoil	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: F
Tian et al. 2019	Huangjiagou watershed, China	1.04	14 years	Uncultivated land, arable land, sandstone	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: F $\delta^{15}\text{N}$: F

(Continues)

TABLE 1 | (Continued)

Study	Location	Watershed (km ²)	Sampling duration	Sediment sources	Tracers	Conc. dependent mixing	Bracket/range test or other test (<i>P</i> =pass; <i>F</i> =fail test)
Sun et al. 2021	Yangtze River Basin, China	1.8×10 ⁶	1 month	Modern biomass, aged-soil carbon, and petrogenic carbon	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: measured (not used)
Studies considering uplands as well as streambanks or floodplains							
Gomez et al. 2003	Waipaoa River basin, New Zealand	1580	11 years	Weathered bedrock from gullies, soil on hillslopes	$\delta^{13}\text{C}$	NO	Performed: NO
Fox and Papanicolaou 2007	Upper Palouse Basin, Idaho, USA	0.71	1 month	Agricultural hillslopes Floodplains	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	NO	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Fox and Papanicolaou 2008a	Upper Palouse Basin, Idaho, USA	15	2 months	Conifer forest, hay pastures, winter wheat and peas	$\delta^{15}\text{N}$	NO	Performed: YES $\delta^{13}\text{C}$: not measured $\delta^{15}\text{N}$: P
Fox and Papanicolaou 2008b	Upper Palouse Basin, Idaho, USA	600	2 months	Winter wheat-uplands, hay pastures, conifer forest	$\delta^{15}\text{N}$	NO	Performed: NO
Rhoton et al. 2008	Walnut Gulch Exp. Watershed, Arizona, USA	150	4 years	Fan remnants, hillslopes, mountain slopes, floodplains	$\delta^{13}\text{C}$	NO	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: not measured
Fox 2009	Surface coal mines, Kentucky, USA	4 (~2 km ²)	6 months	Surface mining, forests, grassland, streambanks	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Gellis et al. 2009	Chesapeake Bay Watershed, USA	109–156	3 years	Construction sites, ditches, agriculture, forested areas	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Mukundan et al. 2010	Southern Piedmont Stream, North Fork Broad River, Georgia, USA	182	6 events	Croplands, pastures, forests, roads, construction, banks	$\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: not measured $\delta^{15}\text{N}$: P
Blake et al. 2012	Furze Brook catchment, River Otter in Devon, UK	1.45	1 event	Maize agriculture, winter wheat agriculture, pastures, wooded areas, banks	$\delta^{13}\text{C}$ CSIA	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: not measured

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TABLE 1 | (Continued)

Study	Location	Watershed (km ²)	Sampling duration	Sediment sources	Tracers	Conc. dependent mixing	Bracket/range test or other test (P =pass; F =fail test)
Jung et al. 2012	Haeen Basin, northern South Korea	60	1 year	Forested areas, cultivated fields, streambanks	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: F $\delta^{15}\text{N}$: F
Gellis and Noe 2013	Linganore Creek, Frederick County, Maryland, USA	147	3 years	Croplands, pastures, forested areas, streambank	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: F
Hancock and Revill 2013	Logan and Albert River, Brisbane, Australia	3860	4 days	Cultivated areas, forested areas, pastures, subsoils, streambanks	$\delta^{13}\text{C}$ CSIA	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: not measured
McKinley, Radcliffe, and Mukundan 2013	North Fork Broad River, Northeast Georgia, USA	182, 50, 33, 7	1 event	Pasture, subsurface, streambanks	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Fox and Martin 2015	Appalachian Forest Region, southeastern Kentucky, USA	3.5, 2.2	3 years	Surface reclaimed mining soils, forested areas, streambanks	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Lacey et al. 2015	Knapp Creek, Upper Bremer River, Blackfellow Creek, Southeast Queensland, Australia	75, 123, 311	2 years	Gullies, cultivated agriculture, grazing pastures, natural grazing land, streambanks	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Stewart, Massoudieh, and Gellis 2015	Laurel Hill Creek watershed, Somerset County, Pennsylvania, USA	324	1 year	Forested areas, roads, pastures, cropland topsoil, streambanks	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Lacey et al. 2016	Mano, Niida, Ota catchments, Japan	171, 265, 77	2 years	Cultivated topsoil, forested areas, subsoils	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Garzon-Garcia et al. 2017	Southeast Queensland, Australia	3076, 75, 2.5	5 months	Gullies, intact valley soil, grass litter, tree litter, streambanks	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Yu and Rhoads 2018	Upper Sangamon River basin, Saybrook, Illinois, USA	84.3	2 years	Cropland, grassland, floodplains	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P

(Continues)

TABLE 1 | (Continued)

Study	Location	Watershed (km ²)	Sampling duration	Sediment sources	Tracers	Conc. dependent mixing	Bracket/range test or other test (P =pass; F =fail test)
Vale et al. 2022	Aroaro stream, south-east Auckland, New Zealand	29	Unspecified	Harvested pine, native forest, pasture, scrub, channel bank, subsoil	$\delta^{13}\text{C}$ CSIA	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: not measured
Lizaga et al. 2024	Barués catchment, Aragón River, Spain	23	1 year	Cropland, Mediterranean forest, pine forest, Highly disturbed areas (subsoil and channel banks)	$\delta^{13}\text{C}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: not measured
Studies considering streambed and other sources							
McConnachie and Petticrew 2006	O'Ne-ail drainage basin, Takla Lake region of northern British Columbia, Canada	75	4 months	Organic matter sources, algae, periphyton, decaying salmon	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$, C/N	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Bonn and Rounds 2010	Tualatin River basin, Oregon, USA	1840	2 years	Leaf litter, upland soils, detritus, suspended sediment, macrophytes, phytoplankton, periphyton, wastewater treatment effluent	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$, C/N	NO	Performed: NO
Fox, Davis, and Martin 2010	Upper South Elkhorn watershed, central Kentucky, USA	61.8	1 month	Streambanks, bank slumping, streambed	$\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: not measured $\delta^{15}\text{N}$: P
Schindler Wildhaber, Liechti, and Alewell 2012	River Enziwigger, Willisau, Canton of Lucerne, Switzerland	31	5 months	Forested areas, pastures, arable land, riverbed	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Cooper et al. 2015	Blackwater subcatchment, River Wensum, Norfolk, UK	5.4	7 months	Organic matter sources, streambed sediments	$\delta^{13}\text{C}$ CSIA	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: not measured
McCorkle et al. 2016	Kings River Experimental Watersheds (KREW), California, USA	Unspecified	5 years	Surface erosion (O horizon), streambanks, streambed	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P

(Continues)

TABLE 1 | (Continued)

Study	Location	Watershed (km ²)	Sampling duration	Sediment sources	Tracers	Conc. dependent mixing	Bracket/range test or other test (<i>P</i> =pass; <i>F</i> =fail test)
Zhang et al. 2017	River Itchen Special Area of Conservation (SAC) in Hampshire, UK	58, 70, 70	9 months	Farmyard manures/slurries, damaged road verges, septic tanks, decaying vegetation, watercress farms, fish farms, bed sediment	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P (1 of 3) $\delta^{15}\text{N}$: P (3 of 3)
Rose, Karwan, and Aufdenkampe 2018	White Clay Creek watershed, southeastern Pennsylvania, USA	7.25	1 year	Agricultural surface soil, forested surface and subsurface soil, pastures, gullies, Streambanks, Floodplains, Streambed	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	NO	Performed: YES $\delta^{13}\text{C}$: F $\delta^{15}\text{N}$: F
Mahoney et al. 2019	Upper South Elkhorn watershed, central Kentucky, USA	61.8	4 years	Upland surfaces Streambed	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: F
Schenk, Harden, and Kelson 2019	Sprague River Basin, south-central Oregon, USA	4167, 1447, 71	2 days	Streambank, Levee, Streambed	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	NO	Performed: YES $\delta^{13}\text{C}$: F $\delta^{15}\text{N}$: P
Valente et al. 2020	Rio Grande do Sul State, southern Brazil	0.83, 1.10	4 years	Eucalyptus plantations, unpaved roads, oats, grassland, Streambed	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: F $\delta^{15}\text{N}$: P
Ford et al. 2020	Fourpole Creek, Huntington, West Virginia, USA	60.6	1 year	River, Streambed	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: F $\delta^{15}\text{N}$: P
Abban et al. 2016	Clear Creek Watershed, southeastern Iowa, USA	26	3 months	Uplands, floodplains, instream sediment, algal and detrital sources	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	NO	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Bahadori et al. 2020	Johnstone River catchment, Australia	2624	2 years	Pastures, sugarcane, forest, banana, Streambed	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P
Guo et al. 2020	Beijing area, China	Unspecified	10 months	Atmospheric C, sewage, algae, soil, C3 and C4 plants	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: F
Xuan et al. 2020	Pearl River Delta, China	7500	2 months	Soil OM, sewage-derived OM, phytoplankton	$\delta^{13}\text{C}$, $\delta^{15}\text{N}$	YES	Performed: YES $\delta^{13}\text{C}$: P $\delta^{15}\text{N}$: P

isotopic ratio conservativeness by optimising the solution space in ensemble modelling. Upadhayay et al. (2021) measured changes in $\delta^{13}\text{C}$ or $\delta^{15}\text{N}$ of soils during transport from their source zones to the edge of field. Cumulatively, these studies measured changes in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ primarily from physical or biological process influence and showed the potential of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of sediment to change during transport and storage.

Riddle et al. (2022) provided the most comprehensive treatment of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ conservativeness, to our knowledge, to date. Riddle et al. (2022) carried out an investigation of processes potentially impacting isotopic ratio conservativeness and other error considerations by studying changes to $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ from carbon oxidation and nitrogen mineralisation, algae accrual into temporary sediment deposits in the river, the physical loss of organic matter via disaggregation, changes occurring during concentration dependent mixing, and time-varying isotopic ratios of sediment sources. The results showed that processes and error considerations can change isotope ratios, however, the net impact of these non-conservative attributes varied widely. Riddle et al. (2022) showed processes impacting conservativeness and other errors could generally be accounted for by following a set of recommendations, such as requiring that each isotopic ratio separates sediment sources by at least 1‰ and carrying out concentration dependent mixing (see Table 1) in (Riddle et al. 2022 for the full list of ten recommendations). The most unresolved process impacting conservativeness in Riddle et al. (2022) was uncertainty associated with changes to $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of the streambed sediment source; and their study called for research to quantify and account for the changes during fingerprinting. For example, their analyses suggested algae accrual in streambed sediment could impact $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ by as much as 8‰ and 2‰, respectively. Riddle et al. (2022) concluded that changes to $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ in streambed deposition zones is likely the largest source of uncertainty hindering usefulness of the isotopic ratios in sediment fingerprinting.

3 | Study Site

We collected and analysed a dataset from a watershed dominated by the streambed sediment source located in central Kentucky, United States. The watershed was the Upper South Elkhorn watershed (65.1 km²) in the inner bluegrass region of Kentucky (Figure 2; Table S1). Headwaters of the South Elkhorn Creek originate in southwestern Lexington, Kentucky, within urban areas and the middle and lower watershed extends into agricultural pastureland. Gently rolling hills and relatively mild slopes characterise the land surface. The stream channel is bedrock-controlled with fine sediment deposits observed throughout the channel length. Our prior work showed sediment connectivity of the uplands approaches only 13% of the watershed surface during the wettest times of the year (Mahoney et al. 2019), and that sediment transported from the bed of the channel constitutes 85% of the sediment collected (Mahoney et al. 2019). This sediment load is primarily fine-sized in composition ($d < 53\ \mu\text{m}$, where d is the aggregate diameter, Fox et al. 2014) owing to the silt loams that primarily make up the South Elkhorn watershed's soil cover. The Upper South Elkhorn watershed is also characterised by long, flat floodplains adjacent to the stream network. Air temperature ranges

between, on average, 0.5°C in January to 24.5°C in July. The average yearly rainfall for this region is 1148 mm. The climate is classified as humid subtropical. The watershed was chosen for analyses due to (i) the dominance of the streambed source reflecting the lack of sediment connectivity with the uplands for many events of the year (Mahoney et al. 2019); (ii) past studies conducted in the watershed that provided an understanding of sediment transport (e.g., Russo and Fox 2012; Mahoney et al. 2019) and (iii) past data collection routine including the establishment of the sediment trap field and laboratory methods to produce the first 8 years of the dataset in this paper, as reported in Ford et al. (2015).

4 | Methods

4.1 | Field Sediment Sample Collection and Laboratory Analyses

The dataset included $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ measurements of sediment collected over 14 years from 2006 to 2020. Sediment samples were collected at two locations in the mainstem of South Elkhorn creek (Figure 2). Sediment trap samples were collected weekly to bi-weekly using in situ sediment traps (Phillips, Russell, and Walling 2000) that accumulate material providing a time-integrated sample of fluvial sediment. For each integrated sample, the sediment trap was collected and emptied to 18.9 L buckets and transported to the laboratory. Samplers were replaced with clean traps that were cleaned using phosphorus free soap and water that was deionised and treated with adsorption to remove organics.

In the laboratory, the buckets with the water-sediment mixture were stored in a refrigerated space for 48 h to allow sedimentation from the water column (Figure S1). Samplers were then decanted, centrifuged, frozen, and freeze-dried to remove remaining water. The bulk sample was sub-sampled depending on mass, wet sieved to retain the less than 53 μm size fraction, brought to a steady state via centrifugation and freeze-drying, and ground to a powder using a Wig-L-Bug Grinding Mill (Ford et al. 2015). Analytical methods were carried out in the University of Arkansas Stable Isotope Laboratory to analyse for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of sediment as well as carbon and nitrogen elemental composition following methods outlined in Riddle et al. (2023). For carbon analyses, powdered samples were weighed into tin capsules and acidified repeatedly with a weak (0.5 M) hydrochloric acid to remove inorganic carbon (carbonate) material. Sediment organic carbon (SOC), sediment nitrogen (SN), $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ were analysed using a Costech 4010 elemental analyser interfaced with a Thermo Finnigan ConFlo III device to a Thermo Finnigan Delta Plus XP isotopic ratio mass spectrometer (IRMS) or a EA-Isolink elemental analyser interfaced (conFlo IV) to a Thermo sci Delta V Advantage. All isotopic results are reported in per mil (‰) relative to V-PDB and air for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively. Average standard deviation for USGS-40 and USGS-41 was 0.06‰ and 0.13‰ for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively. Average standard deviation for the elemental standard (acetanilide) was 0.35‰ and 0.09‰ for SOC and SN, respectively. Average standard deviation for SOC, SN, $\delta^{13}\text{C}$, and $\delta^{15}\text{N}$ of unknowns was 0.07‰, 0.01‰, 0.04‰ and 0.11‰, respectively. For the dissolved nitrate analysis, USGS

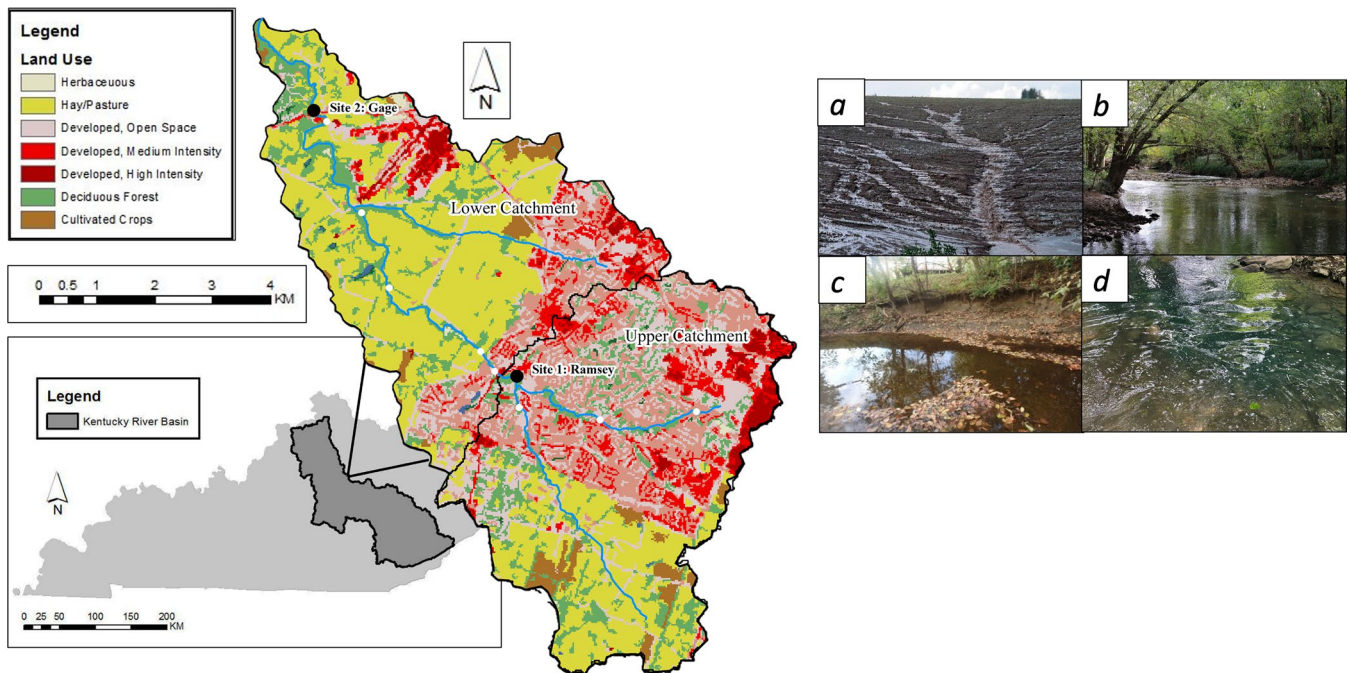


FIGURE 2 | Study watershed, land use, instream sample site locations (from Fox, Davis, and Martin 2010), and stream location within the Kentucky River Basin, USA. Photos of Upper South Elkhorn Creek watershed including (a) ephemeral gully soil erosion, (b) low gradient channel, (c) stream bank cutting and (d) instream algal growth.

32, 34, 35 were used to normalise the raw results with the values of 25‰, −27.9‰, 57.5‰. The average of the standards was 25.67‰, −27.76‰, and 57.46‰ with a standard deviation of 0.29‰, 0.29‰ and 0.23‰ respectively.

4.2 | Empirical Model Decomposition

Empirical mode decomposition (EMD) was employed to quantify the statistical significance in oscillations in SOC, SN, $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ for the 14-year timeseries, expanding upon a previous analysis conducted for the first 8-years of data (see Ford et al. 2015). EMD is an adaptive time-series analysis method that decomposes data into a set of IMFs, in which the lowest frequency IMF is identified as the base (residual) trend and the highest frequency trend is considered noise for well-sampled datasets (Wu et al. 2007). EMD was performed using MATLAB version R2024a. A statistical significance test was performed on EMD results based on the method explained in Wu et al. (2007) to determine if IMFs were significantly different from white noise IMFs. A negative linear relationship of $\log(\text{Var})$ versus $\log(\text{Period})$, with a slope of 1, was plotted with upper and lower bounds for the confidence interval being represented with $\log_{10}(\text{Var}) \pm \log_{10}(3)$. IMF signals that visually plotted near, or outside the specified variance range were determined to be statistically different from white noise and thus have some physical meaning, while those that fell well within the bounds were considered noise. We partitioned the results into significant quasi-seasonal IMFs (summation of significant IMFs with average period ranging from 0.5 to 1.5 years), longer term IMFs (summation of significant IMFs with an average period greater than 1.5 years plus the residual IMF) and noise (non-significant IMFs) (Gerlitz et al. 2023).

4.3 | Mixing Model Analysis

As will be shown in the results, $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of transported sediment and streambed sediment show variability over time, including pronounced seasonality. This seasonal shifting of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ can potentially influence source apportionment results when using the isotopic ratios as sediment tracers. We carried out virtual mixing model simulations using concentration dependent mixing to test the influence of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ seasonality on results as

$$\delta_M^T \rho_M^T = \sum X_i \delta_i^T \rho_i^T \quad (1)$$

where ρ is the elemental concentration; X is the mass fraction; M indicates the mixture location; i indicates source and T indicates the tracer.

Our goal was to test the separation needed from a second sediment source relative to the mean of streambed samples because prior work showed the isotopic ratios of sediment sources should be separated by 1‰ or more (Riddle et al. 2022). We considered additional seasonality of the isotopic ratios would increase this mean separation needed, and the virtual tests provided a means to assess this difference needed (Latorre et al. 2021; Riddle et al. 2022). In the virtual mixing, the two sources were divided equally ($X_i = 0.5$) and mixing was carried forward to produce the mixture results when mean isotopic ratios for both the streambed sediment source was mixed with a second source separated by a fixed per mil value. Thereafter, the seasonality and mean from the statistically significant IMFs was used for the isotopic ratio of the streambed source on the right-hand side of Equation (1) together with the inverse mixing model simulations to reproduce source contributions at each time step. Confidence

bounds were treated as $\pm 15\%$ the correct solution, as mixing modelling contains considerable variance. For each time step, the virtual solutions were tested for correctness within the confidence bounds, and correct simulations were accounted across all time steps. The simulations were repeated for different fixed per mil separation between the sediment sources, for example, 1‰ separation, 2‰ separation, etc.

5 | Results

5.1 | Empirical Mode Decomposition of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ for Sediment

Empirical mode decomposition of the carbon and nitrogen isotopic ratio and elemental measurements of transported sediment show significant long term and seasonal variation of data (Figures 3 and 4). Significant long-term IMFs were present in all parameters measured. The variation of the long-term IMFs produced substantial change in parameter multi-year means, including a net 1.6‰ decrease for $\delta^{13}\text{C}$ from 2006 to 2015 (Figure 3b), 1.1‰ increase for $\delta^{15}\text{N}$ from 2006 to 2014 (Figure 4b), 3.6 gC 100g sed⁻¹ increase for SOC from 2007 to 2015 (Figure 3f), and 0.22 gC 100g sed⁻¹ increase for SN from 2006 to 2015 (Figure 4f).

Empirical model decomposition showed significant IMFs with nearly 1 year sinuosity for all parameters that allowed separation of the isotopic ratio and elemental seasonal variation (Figures 3c and 4c). $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ showed local minima in winter in most years while SOC and SN showed local maxima in late fall and winter in most years. The presence or absence of local minima for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ and maxima for SOC and SN was tabulated for each year of the study period (Table S2). Results showed the late fall and winter extrema were more consistent for carbon measurements as compared to nitrogen. $\delta^{13}\text{C}$ and SOC showed the extrema 70% and 85% of the time while $\delta^{15}\text{N}$ and SN showed the extrema 46% and 70% of the time. We observed the prominent seasonal patterns in the data by calculating monthly means and standard errors for years that showed the presence of the local extrema. Results in Figure 5 show the prominent winter minima centred in December for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$.

The seasonal IMFs decomposed from the isotopic ratio data showed pronounced changes in individual seasons from year to year and moderate changes on average. At times, large shifts from winter minima to summer maxima exceeded 2‰ for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ (Figures 3c and 4c) but, on average, these shifts were 0.6‰ for $\delta^{13}\text{C}$ and 1.0‰ for $\delta^{15}\text{N}$ (Figure 5a,b). Elemental results were also substantial, and shifts for individual years were as high as 5 gC 100g sed⁻¹ for SOC and 0.5 gN 100g sed⁻¹ for SN (Figures 3g and 4g), while average seasonal shifts were 0.6 gC 100g sed⁻¹ and 0.22 gN 100g sed⁻¹ (Figure 5c,d).

Empirical mode decomposition explained more overall data variance for carbon in comparison to nitrogen. Further, EMD explained more of the variance in isotopic data than in the elemental data. 72% of $\delta^{13}\text{C}$ data variance were explained by mean and seasonal IMFs (bold line in Figure 3a) while 28% of data variance were unexplained data variance (Figure 3d). 50% of

$\delta^{15}\text{N}$ data variance were explained by significant IMFs while 50% was unexplained variance (Figure 4a,d). 54% of SOC data variance were explained by significant IMFs while 46% was unexplained variance (Figure 3e,h). 38% of SN data variance were explained by significant IMFs while 62% was unexplained variance (Figure 4e,h).

5.2 | Mixing Model Results

The influence of seasonal variability of the streambed $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ signals on sediment apportionment is shown in Figure 6. Assuming a true streambed fraction of 0.50, the seasonality of the isotopic ratios causes the source fractions to fall outside the confidence bound of this true value for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$. The extent to which the true streambed fraction is misconstrued decreases as the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of the arbitrary second sediment source become more distinct from the streambed, that is the per mil separation grows. Considering $\delta^{13}\text{C}$, a mean shift of 1‰ of the second source from $\delta^{13}\text{C}$ of streambed sediment results in 34% of un-mixing calculations falling within the confidence band selected for source allocation results ($50\% \pm 15\%$), and in turn 66% of calculations were incorrect. A mean shift of 2‰ for the $\delta^{13}\text{C}$ second source results in 71% of calculations falling within the confidence band, and 29% of calculations were incorrect. Mean shifts of 3‰ and 4‰ result in 90% and 97%, respectively, falling within the confidence band for correct results. Considering $\delta^{15}\text{N}$, mean shifts of 1‰, 2‰, 3.6‰ and 4‰ result in 43%, 71%, 90% and 96%, respectively, falling within the confidence band for correct results. The results in Figure 6 reflect mixing model results when each isotopic ratio is simulated independently. We also carried out the mixing model analyses using the isotopic ratios in combination, and the overdetermined tracer matrix was solved with error minimization for all modelling time steps where both tracer mixing values were bracketed by the sources. Results were similar, albeit slightly improved for the relatively small mean shifts in the isotopic ratios (see Figure S2). Mean shifts of 1‰ for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of the second source results in 58% of un-mixing calculations falling within the confidence band selected for source allocation results ($50\% \pm 15\%$), and in turn 42% of calculations were incorrect (Figure S2a). Mean shifts of 2‰, 3‰ and 4‰ for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ result in 80%, 89% and 95%, respectively, falling within the confidence band for correct results (Figure S2b-d).

In summary, mixing model simulations showed that the mean streambed isotopic ratios should be separated from other sediment sources by 3.0‰ and 3.6‰ for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively, when the isotopic ratios are used independently; and the mean streambed value of both isotopic ratios should be separated from other sediment sources by 3.0‰ when the isotopic ratios are used in combination. These separations allowed approximately 90% success in estimating source allocation when considering a continuously changing streambed sediment signature.

5.3 | Checking Robustness of Assumptions

We carried out three assessments to check the robustness of assumptions underlying the present analyses and results of this study: (1) impact of annual maximum flows on $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$; (2)

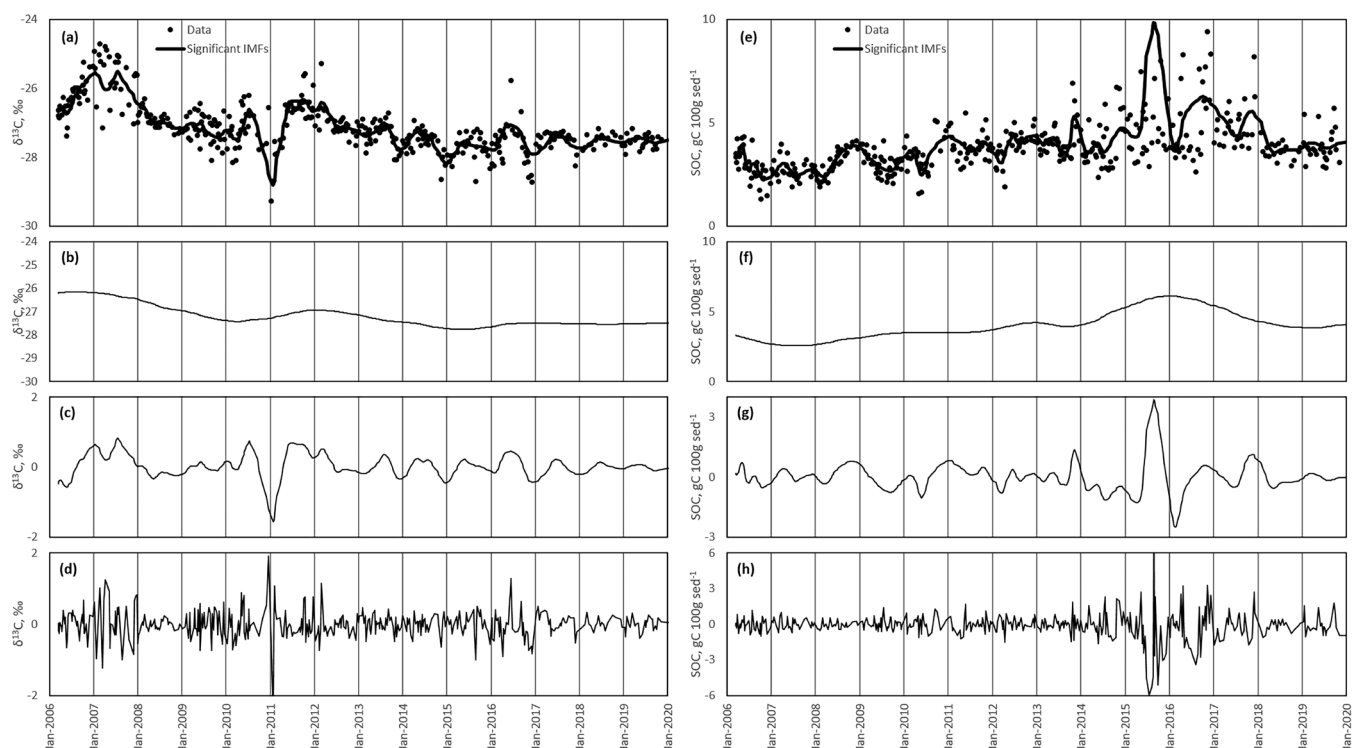


FIGURE 3 | Empirical mode decomposition results for carbon stable isotopic ratios ($\delta^{13}\text{C}$) of sediment and organic carbon concentration of sediment (SOC) including: (a, e) data with significant intrinsic mode functions (IMFs), (b, f) significant long term IMFs, (c, g) significant seasonal IMFs and (d) data noise.

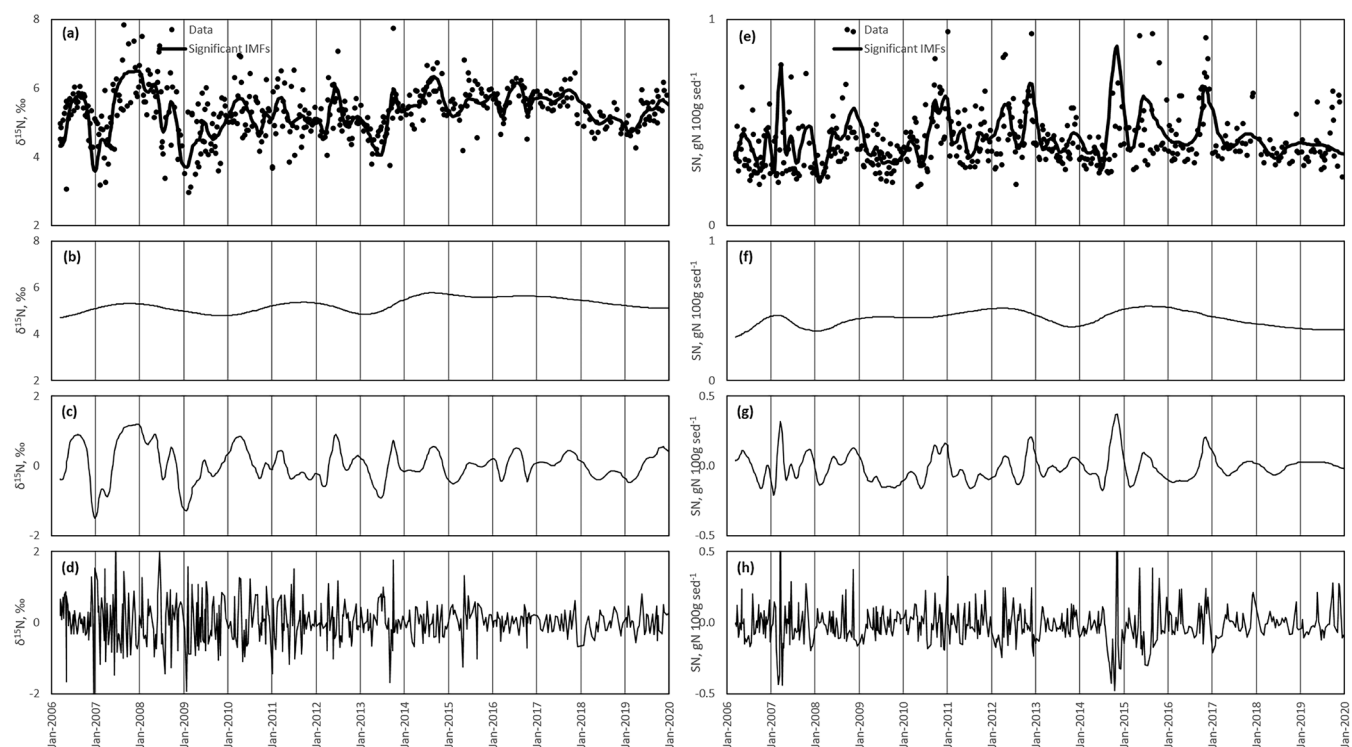


FIGURE 4 | Empirical mode decomposition results for nitrogen stable isotopic ratios ($\delta^{15}\text{N}$) of sediment and nitrogen concentration of sediment (SN) including: (a, e) data with significant intrinsic mode functions (IMFs), (b, f) significant long term IMFs, (c, g) significant seasonal IMFs and (d) data noise.

influence of streambed sediment on EMD results and (3) assessment of seasonality for all sites, not just those with pronounced minima.

We carried out analyses of how $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of the sediment signal was impacted by annual maximums, to investigate the influence of individual storm events that might skew and outweigh

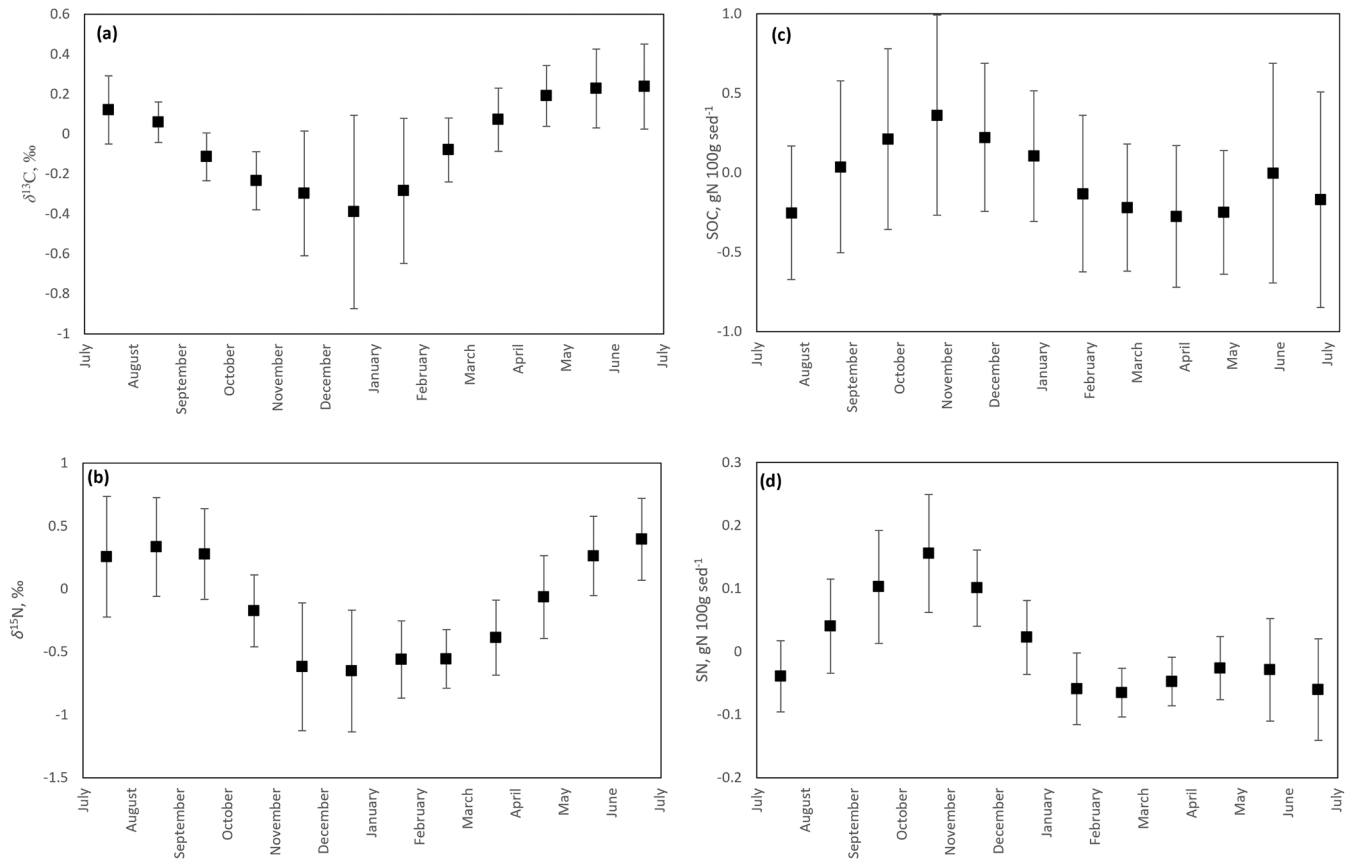


FIGURE 5 | Prominent seasonality of isotopic and elemental measurements including local winter minima for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ and local fall/winter maxima for SOC and SN. Results below reflect of the mean (± 1 SE) of monthly data for years for which the significant seasonal IMFs showed winter minima and maxima, as detailed in Table 1.

the streambed influence. The annual maximum discharge storm event was selected for each year of the study period (Table S3). Two sets of statistical tests were carried out, including one set of tests analysing $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ for data before and after the annual maxima using raw data and a second set of tests analysing $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ before and after the annual maxima using residuals after seasonal detrending was carried out. A change in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ were not statistically significant when comparing datasets before and after the annual maxima (Table S4), which further suggests upland sediment is not controlling the signal. We also analysed data noise of isotopic ratio and elemental values of sediment following EMD (Figures S3 and S4). Symmetry of data noise exhibit normality, and $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ do not exhibit dependence on water flowrate (Q in Figures S2 and S3). Results further suggests storm events are not controlling the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ signal.

We further assessed the dominance of streambed sediment influence on the EMD results for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$. We claim the seasonality of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ resulting from EMD reflects the variation in the streambed source sediment because the dataset is 85% controlled by streambed sediment. To support this claim, we carried out mixing simulations to separate the streambed sediment source from the upland sediment source. The sediment transport numerical modelling results of Mahoney et al. (2019) were used to characterise the allocation of sediment sources as a function of storm event discharge (Figure 9 in Mahoney et al. 2019), and mixing simulation was carried out using the upland sediment source $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values reported for the

basin (Ford et al. 2015; Ford, Fox, and Rowe 2015; Ford, Fox, and Pollock 2017). Empirical model decomposition results for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of the isolated streambed signal shows seasonality very similarly to the transported sediment (Figure S5). In fact, the seasonality is even further supported for $\delta^{13}\text{C}$, and every single year of the $\delta^{13}\text{C}$ seasonal signal from EMD of bed sediment shows the winter minima behaviour.

We also assessed the seasonality of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ results when including all years of the study site, rather than only the years exhibiting local minima (i.e., years designated as ‘yes’ in Table S2). The seasonal trends of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ were similar when including all years, although standard error of results was greater (Figure S6).

6 | Discussion

Results lead to our discussion of processes impacting the conservativeness of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ for streambed sediment as sediment tracers. We also discuss recommendations for their application in sediment fingerprinting.

Seasonal accrual of algae in streambed sediment is hypothesised to result in the observed 0.6‰ and 1.1‰ fluctuations of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively. Four distinct components to this seasonal pattern in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ dynamics are observed: (1) a decrease begins in late summer and fall, (2) a minima occurs

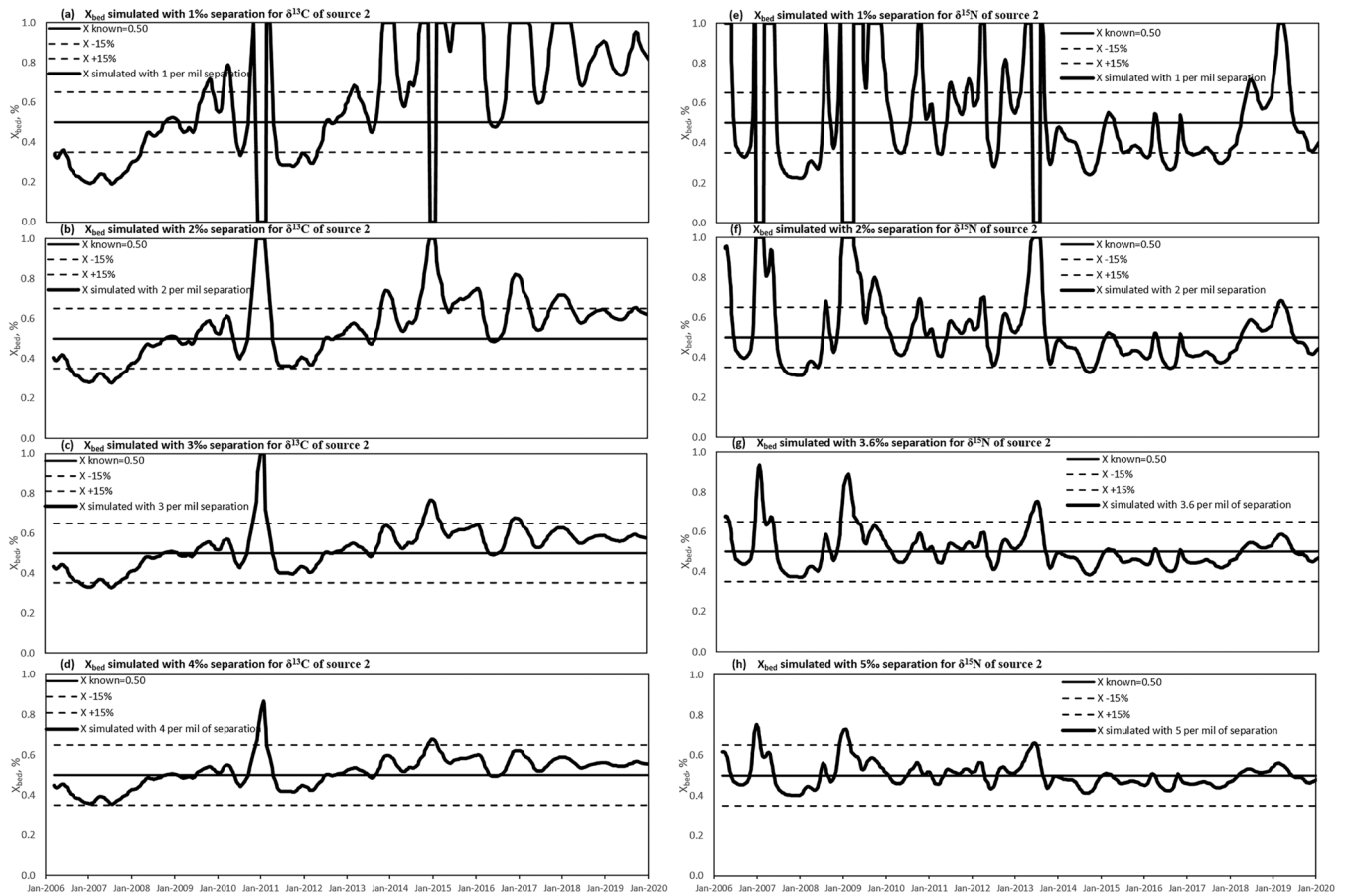


FIGURE 6 | Mixing model results of sediment fraction from the streambed (X_{bed}) when varying isotopic ratio differences between sources.

in early winter, (3) an increase in late winter and spring and (4) a maxima occurs in the summer. First, the streambed promotes organic matter turnover and heterotrophic dominance in late summer and fall. As a result, decreases in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ likely reflect degradation and stabilisation of isotopically lighter algae ($\delta^{13}\text{C} = -37.8 \pm 5.5\text{‰}$ and $\delta^{15}\text{N} = 4.95 \pm 1.6\text{‰}$ for algae) into bed sediment. Prior to algal stabilisation, the bed sediment is composed of material originating from isotopically heavier upland sediment sources, such as banks ($\delta^{13}\text{C} = -25.0 \pm 0.64\text{‰}$ and $\delta^{15}\text{N} = 6.85 \pm 0.51\text{‰}$) and hillslopes ($\delta^{13}\text{C} = -26.55 \pm 0.85\text{‰}$ and $\delta^{15}\text{N} = 5.05 \pm 0.85\text{‰}$) (Ford et al. 2015; Ford and Fox 2015, 2017). As algal stabilisation slows, a minima in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ is reached in the winter. In the spring, increases in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ occur because of resuspension and transport of isotopically lighter bed sediment with algae accrued. Thereafter, deposition of upland sediment to the streambed occurs during late winter and spring hydrologic events.

We dismiss an alternative explanation to our results based on hydrologic data and our prior knowledge of the system. An alternate hypothesis to explain the seasonal patterns in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ is that the fluvial sediment signal is related to hydrologic connectivity of external sources (e.g., Li et al. 2024) rather than dynamic changes within the streambed. That is, material enriched in C-13 atoms and N-15 atoms is delivered to the stream during the summer while material depleted in C-13 and N-15 is delivered in the winter. However, when assessing the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of fluvial sediment during hydrologically well-connected

events, no such dependence of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ on the time-of-year for fluvial sampling is identified (Figure S7). Also, the climate of the watershed does not exhibit a season in which hydrology controls the system, as there is no distinct dry and wet season in central Kentucky (Figure S8a). In turn, results do not exhibit seasonality of water discharge events for this watershed that could lead to a seasonal, hydrologic control on $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ (Figure S8b). While the influence of seasonal hydrology certainly plays a role in determining the isotopic composition of fluvial sediment in many watershed systems, evidence suggests that this impact is lesser than that of streambed biogeochemistry in the South Elkhorn system. This inference is highly consistent with our prior research showing that sediment originating from the streambed dominates the sediment transport events in South Elkhorn creek (Mahoney et al. 2019).

The seasonal results for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of streambed sediment agree well with research from other sites as well as our prior work in the region. Fluvial sediment in bed deposits is recognised to accrue benthic organic matter (Hope, Paterson, and Thrush 2020), and the algae accrual to streambed sediment is referred to as algal stabilisation—a coupled biotic-abiotic process whereby the biomass decomposes to more refractory carbon compounds in aquatic ecosystems (Lara and Thomas 1995; Leloup et al. 2013; Hotchkiss and Hall Jr 2015). Ford, Fox, and Rowe (2015) showed that $\delta^{13}\text{C}$ decreases in value when labile autochthonous organic matter stabilises to sediment. Ford and Fox (2017) carried out benthic ecosystems simulations and

showed that algae subroutines were needed to maintain long term equilibrium of organic matter for the South Elkhorn creek. Their modelling simulations showed algae accrual to bed sediment reached a maximum in early winter, similar to the data results showed here (Figure 5).

Results suggest that $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of the streambed sediment source is continuously changing, and should be accounted in sediment fingerprinting designs using a threshold of 3‰ separation of isotopic ratios for streambed sediment from other sediment sources. We show that 90% accurate estimate of source provenance requires 3.0‰ and 3.6‰ separation for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively, of the streambed source and additional source when the isotopic ratios are used independently; and the mean streambed value of both isotopic ratios should be separated from other sediment sources by 3.0‰ when $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ are used in combination. This far exceeds the 1‰ separation of sources for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ recommended in our earlier work by Riddle et al. (2022). Riddle et al. (2022) recommended that 1‰ separation of sources for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ tracers together with including concentration dependent mixing will sufficiently account for numerous non-conservativeness processes and errors by allowing the tracers to pass the virtual mixtures test. The 1‰ separation and concentration dependent mixing generally accounted for isotopic change during organic matter degradation, physical loss mechanisms impacting isotopic ratios during transit, seasonal changes of isotopic ratios for upland sources and isotopic ratio changes to agricultural source soils due to fertilisation (Table 1) in (Riddle et al. 2022). We update these recommendations to suggest 3‰ separation of isotopic ratios for streambed sediment from other sediment sources when seasonality is to be expected. The 3‰ separation rule of thumb will add confidence to source allocations that uses $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ when the streambed source is expected to contribute substantially to the sediment load.

We point out that the results reported herein are for the South Elkhorn creek—a low-gradient and agriculturally-impacted stream—and we qualify that our recommendations may be limited to systems that behave similarly while other systems may behave differently in terms of seasonality of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$. The South Elkhorn creek is characterised geomorphologically as a low-gradient stream system, which contrasts steep-mountain streams. Low-gradient systems are characterised by strong influence of a biogenic carbon source as opposed to the geogenic source dominance of steep-mountain streams (Ford and Fox 2014a). In this manner, the deposition of fine-sized fluvial sediment ($d < 53\mu\text{m}$, where d is aggregate diameter) occurs in these systems because the energy gradient to transport sediment can be limited. In contrast, steep-mountain streams will have sufficient energy to transport fluvial sediment, and the streambed sediment source is often not significant to the sediment load anyways for example, see sediment fingerprinting with $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ tracers in (Fox 2005; Fox and Martin 2015). Moderately steep systems may show less streambed sediment accrual as compared to low-gradient systems, as well as decreased seasonality of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ given a shorter residence time for temporarily stored sediment.

Sediment fingerprinting studies in low-gradient systems show a prevalence of a fluvial sediment sourced from streambeds, as

reported in many recent papers using $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ (see Table 1). Low-gradient, human-impacted streams are ideal locations for streambeds to influence $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ seasonality as they often have high dissolved nutrient levels and reduced canopy cover, which can expose the benthos to light and promote algal growth (Rider et al. 2024). As agricultural impairment increases, such as through production of corn or other row crops in a watershed, systems may be shifted to even greater algal mat densities (Munn 2018). For this reason, the influence of the streambed on $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ seasonality may be more pronounced in highly agricultural systems.

From a management perspective, the streambed can serve as a temporary store for sediment deposits that originated from outside the stream channel and then sequesters carbon (Ford, Fox, and Pollock 2017). Conservation activities that are implemented in the uplands may not yield immediate improvements in the form of reduced downstream sediment exports. Prior to conservation implementation, upland sediment is delivered to the stream corridor as runoff satisfies the transport capacity of the flow, potentially leaving streambed sediment deposits intact. However, once conservation practises are implemented and the upland source is disconnected, the fluid energy of the stream will be expended to erode the temporary streambed deposits. In this way, the gradual release of streambed storage from the river network may impart a lag time between upland conservation implementation and sediment reduction targets.

7 | Conclusion

The conclusion of this paper are as follows:

1. Hydrologic research reported in the literature shows progress when using carbon and nitrogen stable isotopic ratios ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) in sediment fingerprinting studies, but tracer conservativeness, especially with respect to temporarily stored streambed sediment, remains an open question.
2. Of the 49 studies reviewed that use $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, 86% of studies check conservativeness with a bracket test; 80% of studies carry out concentration dependent mixing; and 33% of studies include streambeds as a sediment source.
3. Results from analysing $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ of fluvial sediment with empirical model decomposition of a 14-year dataset dominated by the streambed sediment source shows significant seasonality of the data; and seasonal and multi-year variance account for 72% and 50% of the total data variation for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively.
4. The prominent seasonality for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ show a mean intra-annual change of 0.6‰ and 1.1‰, and the seasonal change is attributed to algal accrual and organic matter turnover in the streambed sediment deposits.
5. Mixing model simulations show that the mean streambed isotopic ratios should be separated from other sediment sources by 3.0‰ and 3.6‰ for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, respectively, to achieve 90% accuracy in source apportionment when the isotopic ratios are used independently; and the mean streambed value of both isotopic ratios should be separated

from other sediment sources by 3.0‰ when $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ are used in combination.

6. Our results lead to the recommendation that mean isotope ratios of sources be separated by at least 3‰ when the streambed is expected to be a prominent sediment source, which far exceeds the prior recommendation of 1‰ mean separation of sources.

Acknowledgements

We thank the numerous undergraduate and graduate students who assisted with collecting the sediment trap samples in the field and processed the samples in the sediment laboratory the past 20 years. We gratefully acknowledge the financial support of this research from the Kentucky Senate Bill 271B Water Quality program and National Science Foundation Awards #1632888, #2119853 and #1933779.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.